24.5. The infinitely repeated economy

Consider repeating our one-period economy forever. At each $t \geq 1$, each household chooses $\xi_t \in X$, with the result that the average $x_t \in X$; the government chooses $y_t \in Y$. We use the notation $(\vec{x}, \vec{y}) = \{(x_t, y_t)\}_{t=1}^{\infty}$, $\vec{\xi} = \{\xi_t\}_{t=1}^{\infty}$. To denote a *history* of (x_t, y_t) up to t, we use the notation $x^t = \{x_s\}_{s=1}^{t}$, $y^t = \{y_s\}_{s=1}^{t}$. These histories live in the spaces X^t and Y^t, respectively, where $X^t = X \times \cdots \times X$, the Cartesian product of X taken t times, and Y^t is the Cartesian product of Y taken t times.[9]

For the repeated economy, the government evaluates paths $(\vec{x}, \vec{y})$ according to

$$V_g(\vec{x}, \vec{y}) = \frac{(1-\delta)}{\delta} \sum_{t=1}^{\infty} \delta^t \, r(x_t, y_t), \tag{24.5.1}$$

where $r(x_t, y_t) \equiv u(x_t, x_t, y_t)$ and $0 < \delta < 1$.[10] A *pure strategy* is defined as a sequence of functions, the tth element of which maps the history (x^{t-1}, y^{t-1}) observed at the beginning of t into an action at t. In particular, for the aggregate of households, a strategy is a sequence $\sigma^h = \{\sigma_t^h\}_{t=1}^{\infty}$ such that

$$\begin{aligned} &\sigma_1^h \in X \\ &\sigma_t^h : X^{t-1} \times Y^{t-1} \to X \qquad \text{for each} \quad t \geq 2 \ . \end{aligned}$$

Similarly, for the government, a strategy $\sigma^g = \{\sigma_t^g\}_{t=1}^{\infty}$ is a sequence such that

$$\begin{aligned} &\sigma_1^g \in Y \\ &\sigma_t^g : X^{t-1} \times Y^{t-1} \to Y \qquad \text{for each} \quad t \geq 2. \end{aligned}$$

We call $\sigma = (\sigma^h, \sigma^g)$ a *strategy profile*. We let $\sigma_t = (\sigma_t^h, \sigma_t^g)$ be the tth component of the strategy profile.

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[9] Marco Bassetto's work (2002, 2005) shows that this specification, which is common in the literature, excludes some interesting applications. In particular, it rules out contexts in which the set of time t actions available to the government is influenced by past actions taken by households. Such excluded examples prevail, for example, in the fiscal theory of the price level. To construct sustainable plans in those interesting environments, Bassetto (2002, 2005) refines the notion of sustainability to include a more complete theory of the government's behavior off an equilibrium path.

[10] Note that we have not defined the government's payoff when $\xi_t \neq x_t$. See footnote 7.

24.5.1. A strategy profile implies a history and a value

APS begin with the insight that a strategy profile $\sigma = (\sigma^g, \sigma^h)$ evidently recursively generates a trajectory of outcomes $\{[x(\sigma)_t, y(\sigma)_t]\}_{t=1}^{\infty}$:

$$\begin{aligned}\big[x(\sigma)_1, y(\sigma)_1\big] &= (\sigma_1^h, \sigma_1^g)\\ \big[x(\sigma)_t,\ y(\sigma)_t\big] &= \sigma_t\big[x(\sigma)^{t-1},\ y(\sigma)^{t-1}\big].\end{aligned}$$

Therefore, a strategy profile also generates a pair of values for the government and the representative private agent. In particular, the value for the government of a strategy profile $\sigma = (\sigma^h, \sigma^g)$ is the value of the trajectory that it generates

$$V_g(\sigma) = V_g\big[\vec{x}(\sigma),\ \vec{y}(\sigma)\big].$$

24.5.2. Recursive formulation

A key step toward APS's recursive formulation comes from defining *continuation stategies* and their associated *continuation values*. Since the value of a path $(\vec{x}, \vec{y})$ in equation (24.5.1) is additively separable in its one-period returns, we can express the value recursively in terms of a one-period economy and a continuation economy. In particular, the value to the government of an outcome sequence $(\vec{x}, \vec{y})$ can be represented

$$V_g(\vec{x}, \vec{y}) = (1-\delta)\, r(x_1, y_1) + \delta V_g\big(\{x_t\}_{t=2}^{\infty},\ \{y_t\}_{t=2}^{\infty}\big) \tag{24.5.2}$$

and the value for a household can also be represented recursively. Notice how a strategy profile σ induces a strategy profile for the continuation economy, as follows. Let $\sigma|_{(x^t,y^t)}$ denote the strategy profile for a continuation economy whose first period is $t+1$ and that is initiated after history (x^t, y^t) has been observed; here $(\sigma|_{(x^t,y^t)})_s$ is the sth component of $(\sigma|_{(x^t,y^t)})$, which for $s \geq 2$ is a function that maps $X^{s-1} \times Y^{s-1}$ into $X \times Y$, and for $s = 1$ is a point in $X \times Y$. Thus, after a first-period outcome pair (x_1, y_1), strategy σ induces the continuation strategy

$$\begin{aligned}(\sigma|_{(x_1,y_1)})_{s+1}\,(\nu^s, \eta^s) &= \sigma_{s+2}\,(x_1, \nu_1, \ldots, \nu_s, y_1, \eta_1, \ldots, \eta_s)\\ &\qquad \text{for all } \ (\nu^s, \eta^s) \in X^s \times Y^s\ , \quad \forall s \geq 0.\end{aligned}$$

It might be helpful to write out a few terms for $s = 0, 1, \ldots$:

$$\begin{aligned}(\sigma|_{(x_1,y_1)})_1 &= \sigma_2(x_1, y_1) = (\nu_1, \eta_1)\\ (\sigma|_{(x_1,y_1)})_2(\nu_1, \eta_1) &= \sigma_3(x_1, \nu_1, y_1, \eta_1) = (\nu_2, \eta_2)\\ (\sigma|_{(x_1,y_1)})_3(\nu_1, \nu_2, \eta_1, \eta_2) &= \sigma_4(x_1, \nu_1, \nu_2, y_1, \eta_1, \eta_2) = (\nu_3, \eta_3).\end{aligned}$$

More generally, define the continuation strategy

$$\begin{aligned}(\sigma|_{(x^t,y^t)})_1 =& \sigma_{t+1}(x^t, y^t)\\ (\sigma|_{(x^t,y^t)})_{s+1}(\nu^s, \eta^s) =& \sigma_{t+s+1}\left(x_1, \ldots, x_t, \nu_1, \ldots, \nu_s; y_1, \ldots, y_t, \eta_1, \ldots, \eta_s\right)\\ &\text{for all } \ s \geq 1 \quad \text{and all} \quad (\nu^s, \eta^s) \in X^s \times Y^s \ .\end{aligned}$$

Here $(\sigma|_{(x^t,y^t)})_{s+1}\ (\nu^s, \eta^s)$ is the induced strategy pair to apply in the $(s+1)$th period of the continuation economy. We attain this strategy by shifting the original strategy forward t periods and evaluating it at history $(x_1, \ldots, x_t, \nu_1, \ldots, \nu_s; y_1, \ldots, y_t, \eta_1, \ldots, \eta_s)$ for the *original* economy.

In terms of the continuation strategy $\sigma|_{(x_1,y_1)}$, from equation (24.5.2) we know that $V_g(\sigma)$ can be represented as

$$V_g(\sigma) = (1-\delta)r(x_1, y_1) + \delta V_g(\sigma|_{(x_1,y_1)}). \tag{24.5.3}$$

Representation (24.5.3) decomposes the value to the government of strategy profile σ into a one-period return and the continuation value $V_g(\sigma|_{(x_1,y_1)})$ associated with the continuation strategy $\sigma|_{(x_1,y_1)}$.

Any sequence $(\vec{x}, \vec{y})$ in equation (24.5.2) or any strategy profile σ in equation (24.5.3) can be assigned a value. We want a notion of an *equilibrium* strategy profile.

24.6. Subgame perfect equilibrium (SPE)

Definition 6: A strategy profile $\sigma = (\sigma^h, \sigma^g)$ is a *subgame perfect equilibrium* (SPE) of the infinitely repeated economy if for each $t \geq 1$ and each history $(x^{t-1},\, y^{t-1}) \in X^{t-1} \times Y^{t-1}$

(a) The private sector outcome $x_t = \sigma_t^h\,(x^{t-1}, y^{t-1})$ is consistent with competitive equilibrium when $y_t = \sigma_t^g\,(x^{t-1}, y^{t-1})$;

(b) For each possible government action $\eta \in Y$

$$(1-\delta)r(x_t, y_t) + \delta V_g(\sigma|_{(x^t, y^t)}) \geq (1-\delta)\, r(x_t, \eta) + \delta V_g(\sigma|_{(x^t; y^{t-1}, \eta)}).$$

Requirement (a) says two things. It attributes a theory of forecasting government behavior to members of the public, in particular, that they use the time t component σ_t^g of the government's strategy and information available at the end of period $t-1$ to forecast the government's decision at t. Condition (a) also asserts that a competitive equilibrium appropriate to the public's forecast value for y_t is the outcome at time t. Requirement (b) says that at each point in time and following each history, the government has no incentive to deviate from the first-period action called for by its strategy σ^g; that is, the government always wants to choose as the public expects. Notice how in condition (b), the government *contemplates* setting its time t choice η_t at something other than the value forecast by the public, but confronts consequences that deter it from choosing an η_t that fails to confirm the public's expectations of it.

In section 24.15 we'll discuss the following question: who *chooses* σ^g, the government or the public? This question arises naturally because σ^g is *both* the government's sequence of policy functions *and* the private sector's rule for forecasting government behavior. Condition b of definition 6 says that the government chooses to confirm the public's forecasts.

Definition 6 implies that for each $t \geq 2$ and each $(x^{t-1}, y^{t-1}) \in X^{t-1} \times Y^{t-1}$, the continuation strategy $\sigma|_{(x^{t-1}, y^{t-1})}$ is itself an SPE. We state this formally for $t = 2$.

Proposition 1: Assume that σ is an SPE. Then for all $(\nu, \eta) \in X \times Y$, $\sigma|_{(\nu,\eta)}$ is an SPE.

Proof: Write out requirements a and b that Definition 6 asserts that the continuation strategy $\sigma|_{(\nu,\eta)}$ must satisfy to qualify as an SPE. In particular,

for all $s \geq 1$ and for all $(x^{s-1}, y^{s-1}) \in X^{s-1} \times Y^{s-1}$, we require

$$(x_s, y_s) \in C, \tag{24.6.1}$$

where $x_s = \sigma^h|_{(\nu,\eta)}(x^{s-1}, y^{s-1}), y_s = \sigma^g|_{(\nu,\eta)}(x^{s-1}, y^{s-1})$. We also require that for all $\tilde{\eta} \in Y$,

$$(1-\delta)r(x_s, y_s) + \delta V_g(\sigma|_{(\eta,x^s;\nu,y^s)}) \geq (1-\delta)r(x_s, \tilde{\eta}) + \delta V_g(\sigma|_{(\nu,x^s;\eta,y^{s-1},\tilde{\eta})}) \tag{24.6.2}$$

Notice that requirements a and b of Definition 6 for $t = 2, 3, \ldots$ imply expressions (24.6.1) and (24.6.2) for $s = 1, 2, \ldots$. ∎

The statement that $\sigma|_{(\nu,\eta)}$ is an SPE for all $(\nu, \eta) \in X \times Y$ ensures that σ is *almost* an SPE. If we know that $\sigma|_{(\nu,\eta)}$ is an SPE for all $(\nu, \eta) \in (X \times Y)$, we must only add two requirements to ensure that σ is an SPE: first, that the $t = 1$ outcome pair (x_1, y_1) is a competitive equilibrium, and second, that the government's choice of y_1 satisfies the time 1 version of the incentive constraint b in Definition 6.

This reasoning leads to the following lemma that is at the heart of the APS analysis:

Lemma: Consider a strategy profile σ, and let the associated first-period outcome be given by $x = \sigma_1^h, y = \sigma_1^g$. The profile σ is an SPE if and only if

(1) for each $(\nu, \eta) \in X \times Y$, $\sigma|_{(\nu,\eta)}$ is an SPE;

(2) (x, y) is a competitive equilibrium;

(3) $\forall\ \eta \in Y$, $(1-\delta)\, r(x, y) + \delta\, V_g(\sigma|_{(x,y)}) \geq (1-\delta)\, r(x, \eta) + \delta V_g(\sigma|_{(x,\eta)})$.

Proof: First, prove the "if" part. Property a of the lemma and properties (24.6.1) and (24.6.2) of Proposition 1 show that requirements a and b of Definition 6 are satisfied for $t \geq 2$. Properties (2) and (3) of the lemma imply that requirements a and b of Definition 6 hold for $t = 1$.

Second, prove the "only if" part. Part (1) of the lemma follows from Proposition 1. Parts (2) and (3) of the lemma follow from requirements a and b of Definition 6 for $t = 1$. ∎

The lemma is very important because it characterizes SPEs in terms of a first-period competitive equilibrium outcome pair (x, y), and a *pair* of continuation values: a value $V_g(\sigma|_{(x,y)})$ to be awarded to the government next period

if it adheres to the y component of the first-period pair (x, y), and a value $V_g(\sigma|_{(x,\eta)})$, $\eta \neq y$, to be awarded to the government if it deviates from the expected y component. Each of these values has to be selected from a set of values $V_g(\sigma)$ that are associated with some SPE σ.

24.7. Examples of SPE

24.7.1. Infinite repetition of one-period Nash equilibrium

It is easy to verify that the following strategy profile $\sigma^N = (\sigma^h, \sigma^g)$ forms an SPE: $\sigma_1^h = x^N, \sigma_1^g = y^N$ and for $t \geq 2$

$$\begin{aligned} \sigma_t^h &= x^N \qquad \forall\ t\ , \quad \forall\ (x^{t-1}, y^{t-1}); \\ \sigma_t^g &= y^N \qquad \forall\ t\ , \quad \forall\ (x^{t-1}, y^{t-1}). \end{aligned}$$

These strategies instruct the households and the government to choose the static Nash equilibrium outcomes for all periods for all histories. Evidently, for these strategies, $V_g(\sigma^N) = v^N = r(x^N, y^N)$. Furthermore, for these strategies the continuation value $V_g(\sigma|_{(x^t;y^{t-1},\eta)}) = v^N$ for all outcomes $\eta \in Y$. These strategies satisfy requirement a of Definition 6 because (x^N, y^N) is a competitive equilibrium. The strategies satisfy requirement b because $r(x^N, y^N) = \max_{y \in Y} r(x^N, y)$ and because the continuation value $V_g(\sigma) = v^N$ is independent of the action chosen by the government in the first period. In this SPE, $\sigma_t^N = \{\sigma_t^h, \sigma_t^g\} = (x^N, y^N)$ for all t and for all (x^{t-1}, y^{t-1}), and value $V_g(\sigma^N)$ and continuation values $V_g(\sigma^N|_{(x^t,y^t)})$ for each history (x^t, y^t) equal v^N.

It is useful to think about this SPE in terms of the lemma. To verify that σ^N is a SPE, we work with the first-period outcome pair (x^N, y^N) and the pair of values $V_g(\sigma|_{(x^N,y^N)}) = v^N, V_g(\sigma|_{(x,\eta)}) = v^N$, where $v^N = r(x^N, y^N)$. With these settings, we can verify that (x^N, y^N) and v^N satisfy requirements (1), (2), and (3) of the lemma.

24.7.2. Supporting better outcomes with trigger strategies

The public can have a system of expectations about the government's behavior that induces the government to choose a better-than-Nash outcome $(\tilde{x}, \tilde{y}) \in C$. Thus, suppose that the public expects that so long as the government chooses $\tilde{y}$, it will continue to do so in the future, but if ever the government deviates from this choice, thereafter the public expects that the government will choose y^N, prompting the public (or what we can call "the market") to react with $x^N = h(y^N)$. This system of expectations confronts the government with the prospect of being "punished by the market's expectations" if it chooses to deviate from $\tilde{y}$.

To formalize this idea, we shall use the SPE σ^N as a continuation strategy and the value v^N as a continuation value on the right side of part (b) of Definition 6 of an SPE (for $\eta \neq y_t$); then by working backward one step, we shall try to construct *another* SPE $\tilde{\sigma}$ with first-period outcome $(\tilde{x}, \tilde{y}) \neq (x^N, y^N)$. In particular, for our new SPE $\tilde{\sigma}$ we propose to set

$$\begin{aligned} \tilde{\sigma}_1 &= (\tilde{x}, \tilde{y}) \\ \tilde{\sigma}|_{(x,y)} &= \begin{cases} \tilde{\sigma} & \text{if } (x,y) = (\tilde{x}, \tilde{y}) \\ \sigma^N & \text{if } (x,y) \neq (\tilde{x}, \tilde{y}) \end{cases} \end{aligned} \tag{24.7.1}$$

where $(\tilde{x}, \tilde{y}) \in C$ is a competitive equilibrium that satisfies the following particular case of part b of Definition 6:

$$\tilde{v} = (1-\delta)\, r(\tilde{x}, \tilde{y}) + \delta \tilde{v} \geq (1-\delta)\, r(\tilde{x}, \eta) + \delta v^N\,, \tag{24.7.2}$$

for all $\eta \in Y$. Inequality (24.7.2) is equivalent with

$$\max_{\eta \in Y} r(\tilde{x}, \eta) - r(\tilde{x}, \tilde{y}) \leq \frac{\delta}{1-\delta}\,(\tilde{v} - v^N). \tag{24.7.3}$$

For any $(\tilde{x}, \tilde{y}) \in C$ that satisfies expression (24.7.3) with $\tilde{v} = r(\tilde{x}, \tilde{y})$, strategy (24.7.1) is an SPE with value $\tilde{v}$.

If $(\tilde{x}, \tilde{y}) = (x^R, y^R)$ satisfies inequality (24.7.3) with $\tilde{v} = r(x^R, y^R)$, then repetition of the Ramsey outcome (x^R, y^R) is supportable by a subgame perfect equilibrium of the form (24.7.1).

This construction uses the following objects:[11]

1. A proposed first-period competitive equilibrium $(\tilde{x}, \tilde{y}) \in C$.
2. An SPE σ^2 with value $V_g(\sigma^2)$ that is used as the continuation strategy in the event that the first-period outcome does not equal $(\tilde{x}, \tilde{y})$, so that $\tilde{\sigma}|_{(x,y)} = \sigma^2$, if $(x, y) \neq (\tilde{x}, \tilde{y})$. In the example, $\sigma^2 = \sigma^N$ and $V_g(\sigma^2) = v^N$.
3. An SPE σ^1, with value $V_g(\sigma^1)$, used to define the continuation value to be assigned after first-period outcome $(\tilde{x}, \tilde{y})$; and an associated continuation strategy $\tilde{\sigma}|_{(\tilde{x},\tilde{y})} = \sigma^1$. In the example, $\sigma^1 = \tilde{\sigma}$, which is defined recursively (and self-referentially) via equation (24.7.1).
4. A candidate for a new SPE $\tilde{\sigma}$ and a corresponding value $V_g(\tilde{\sigma})$. In the example, $V_g(\tilde{\sigma}) = r(\tilde{x}, \tilde{y})$.

Note how we have used the lemma in verifying that $\tilde{\sigma}$ is an SPE. We start with the SPE σ^N with associated value v^N. We guess a first-period outcome pair $(\tilde{x}, \tilde{y})$ and a value $\tilde{v}$ for a new SPE, where $\tilde{v} = r(\tilde{x}, \tilde{y})$. Then we verify requirements (2) and (3) of the lemma with $(\tilde{v}, v^N)$ as continuation values and $(\tilde{x}, \tilde{y})$ as first-period outcomes.

24.7.3. When reversion to Nash is not bad enough

For discount factors δ sufficiently close to one, it is typically possible to support repetition of the Ramsey outcome (x^R, y^R) with a section 24.7.2 trigger strategy of form (24.7.1). This finding conforms with a version of the folk theorem about repeated games. However, there exist discount factors δ so small that the continuation value associated with infinite repetition of the one-period Nash outcome is not low enough to support repetition of Ramsey. Anticipating that it will revert to repetition of Nash after a deviation then can at best support a lower value for the government that than that associated with repetition of Ramsey outcome, although perhaps its is better than repeating the Nash outcome.

In this circumstance is there a better SPE than can be supported by anticipating version to repetition of the one-period Nash outcome? To support something *better* evidently requires finding an SPE that has a value *worse* than that associated with repetition of the one-period Nash outcome. Following APS, we shall soon see that the best and worst equilibrium values are linked.

[11] In the example, objects 3 and 4 are equated.

24.8. Values of all SPEs

The role played by the lemma in analyzing our two examples hints at the central role that it plays in methods that APS developed for describing and computing values for *all* the subgame perfect equilibria. APS build on the way that the lemma characterizes SPE values in terms of a first-period competitive equilibrium outcome, along with a pair of continuation values, each element of which is itself a value associated with some SPE. The lemma directs APS's attention away from a *set of strategy profiles* σ and toward a *set of values* $V_g(\sigma)$ associated with those profiles. They define the set V of values associated with subgame perfect equilibria:

$$V = \{V_g(\sigma) \mid \sigma \text{ is an SPE}\}.$$

Evidently, $V \subset I\!R$. From the lemma, for a given competitive equilibrium $(x, y) \in C$, there exists an SPE σ for which $x = \sigma_1^h, y = \sigma_1^g$ if and only if there exist two values $(v_1, v_2) \in V \times V$ such that

$$(1-\delta)\, r(x,y) + \delta v_1 \geq (1-\delta)\, r(x,\eta) + \delta v_2 \quad \forall\, \eta \in Y. \tag{24.8.1}$$

Let σ^1 and σ^2 be subgame perfect equilibria for which $v_1 = V_g(\sigma^1)$, $v_2 = V_g(\sigma^2)$. The SPE σ that supports $(x, y) = (\sigma_1^h, \sigma_1^g)$ is completed by specifying the continuation strategies $\sigma|_{(x,y)} = \sigma^1$ and $\sigma|_{(\nu,\eta)} = \sigma^2$ if $(\nu, \eta) \neq (x, y)$.

This construction uses two continuation values $(v_1, v_2) \in V \times V$ to create an SPE σ with value $v \in V$ given by

$$v = (1-\delta)\, r(x,y) + \delta v_1 \ .$$

Thus, the construction maps *pairs* of continuation values (v_1, v_2) into a strategy profile σ with first-period competitive equilibrium outcome (x, y) and a value $v = V_g(\sigma)$.

APS characterize subgame perfect equilibria by studying a mapping from pairs of continuation values $(v_1, v_2) \in V \times V$ into values $v \in V$. They use the following definitions:

Definition 7: Let $W \subset I\!R$. A 4-tuple (x, y, w_1, w_2) is said to be *admissible with respect to* W if $(x, y) \in C$, $(w_1, w_2) \in W \times W$, and

$$(1-\delta)\, r(x,y) + \delta w_1 \geq (1-\delta)\, r(x,\eta) + \delta w_2\,, \quad \forall\, \eta \in Y. \tag{24.8.2}$$

Notice that when $W \subset V$, the admissible 4-tuple (x, y, w_1, w_2) determines an SPE with strategy profile

$$\sigma_1 = (x, y),\ \sigma|_{(x,y)} = \sigma^1,\ \sigma|_{(\nu,\eta)} = \sigma^2 \ \text{ for } (\nu, \eta) \neq (x, y)$$

where σ_1 is a continuation strategy that yields value $w_1 = V_g(\sigma^1)$ and σ_2 is a strategy that yields continuation value $w_2 = V_g(\sigma^2)$. The value of the SPE is $V_g(\sigma) = w = (1-\delta)\, r(x, y) + \delta w_1$.

We want to find the set V.

24.8.1. Basic idea of dynamic programming squared

In Definition 7, W serves as a set of candidate continuation values. The idea is to pick an $(x, y) \in C$, then to check whether we can find $(w_1, w_2) \in W \times W$ that would make the government want to adhere to the y component if w_1 and w_2 could be used as continuation values for adhering to and deviating from y, respectively. If the answer is yes, we say that the 4-tuple (x, y, w_1, w_2) is "admissible with respect to W". Because we have verified that the incentive constraints are satisfied, a yes answer allows us to calculate the *value* (i.e., the left side of (24.8.2)) that can be supported with w_1, w_2 as continuation values. Thus, the idea is to use (24.8.2) to define a mapping from values tomorrow to values today, like that used in dynamic programming. In the next section, we'll define $B(W)$ as the set of possible values attained with admissible pairs of continuation values drawn from $W \times W$. Then we'll view B as an operator mapping sets of continuation values W into sets of values $B(W)$. This operator is the counterpart to the T operator associated with ordinary dynamic programming.

To pursue this analogy, recall the Bellman equation associated with the McCall model of chapter 6:

$$Q = \int \max\left\{\frac{w}{1-\beta}, c + \beta Q\right\} d\,F(w).$$

Here $Q \in I\!R$ is the expected discounted value of an unemployed worker's income *before* he has drawn a wage offer. The right side defines an operator $T(Q)$, so that the Bellman equation is

$$Q = T(Q). \tag{24.8.3}$$

Just as the right side of (24.8.3) takes a candidate continuation value Q for tomorrow and maps it into a value $T(Q)$ for today, APS define a mapping $B(W)$ that, by considering only admissible 4-tuples, maps a *set* of values W tomorrow into a new *set* $B(W)$ of values today. Thus, APS use admissible 4-tuples to map candidate continuation values tomorrow into new candidate values today. In the next section, we'll iterate to convergence on $B(W)$, but as we'll see, it won't work to start from just any initial set W. We have to start from a big enough set.

24.9. APS machinery

Definition 8: For each set $W \subset I\!R$, let $B(W)$ be the set of possible values $w = (1-\delta)\, r(x,y) + \delta w_1$ associated with admissible tuples (x, y, w_1, w_2).

Think of W as a set of potential continuation values and $B(W)$ as the set of values that they support. From the definition of admissibility it immediately follows that the operator B is *monotone*.

Property (monotonicity of B): If $W \subseteq W' \subseteq R$, then $B(W) \subseteq B(W')$.

Proof: It can be verified directly from the definition of admissible 4-tuples that if $w \in B(W)$, then $w \in B(W')$: simply use the (w_1, w_2) pair that supports $w \in B(W)$ to support $w \in B(W')$. ∎

It can also be verified that $B(\cdot)$ maps compact sets W into compact sets $B(W)$.

The self-supporting character of subgame perfect equilibria is referred to in the following definition:

Definition 9: The set W is said to be *self-generating* if $W \subseteq B(W)$.

Thus, a set of continuation values W is said to be self-generating if it is contained in the set of *values* $B(W)$ that are generated by pairs of continuation values selected from W. This description makes us suspect that if a set of values is self-generating, it must be a set of SPE values. Indeed, notice that by virtue of the lemma, the set V of SPE values $V_g(\sigma)$ is self-generating. Thus, we can

write $V \subseteq B(V)$. APS show that V is the *largest* self-generating set. The key to showing this point is the following theorem:[12]

THEOREM 1 (A self-generating set is a subset of V): If $W \subset I\!R$ is bounded and self-generating, then $B(W) \subseteq V$.

The proof is based on "forward induction" and proceeds by taking a point $w \in B(W)$ and constructing an SPE with value w.

PROOF: Assume $W \subseteq B(W)$. Choose an element $w \in B(W)$ and transform it as follows into a subgame perfect equilibrium:

Step 1. Because $w \in B(W)$, we know that there exist outcomes (x, y) and values w_1 and w_2 that satisfy

$$\begin{aligned} w = (1-\delta)\, r(x,y) + \delta w_1 &\geq (1-\delta)\, r(x,\eta) + \delta w_2 \quad \forall \eta \in Y \\ (x,y) &\in C \\ w_1, w_2 &\in W \times W. \end{aligned}$$

Set $\sigma_1 = (x, y)$.

Step 2. Since $w_1 \in W \subseteq B(W)$, there exist outcomes $(\tilde{x}, \tilde{y})$ and values $(\tilde{w}_1, \tilde{w}_2) \in W$ that satisfy

$$\begin{aligned} w_1 = (1-\delta)\, r(\tilde{x},\tilde{y}) + \delta \tilde{w}_1 &\geq (1-\delta)\, r(\tilde{x},\eta) + \delta \tilde{w}_2, \quad \forall\ \eta \in Y \\ (\tilde{x},\tilde{y}) &\in C. \end{aligned}$$

Set the first-period outcome in period 2 (the outcome to occur *given* that y was chosen in period 1) equal to $(\tilde{x}, \tilde{y})$; that is, set $(\sigma|_{(x,y)})_1 = (\tilde{x}, \tilde{y})$.

Continuing in this way, for each $w \in B(W)$, we can create a sequence of continuation values $w_1, \tilde{w}_1, \tilde{\tilde{w}}_1, \ldots$ and a corresponding sequence of first-period outcomes (x, y), $(\tilde{x}, \tilde{y})$, $(\tilde{\tilde{x}}, \tilde{\tilde{y}})$.

At each stage in this construction, policies are *unimprovable*, which means that given the continuation values, one-period deviations from the prescribed policies are not optimal. It follows that the strategy profile is optimal. By construction $V_g(\sigma) = w$. ∎

[12] The *unbounded* set $I\!R$ (the extended real line) is self-generating but not meaningful. It is self-generating because any value $v \in I\!R$ can be supported if there are no limits on the continuation values. It is not meaningful because most points in $I\!R$ are values that cannot be attained with *any* strategy profile.

Collecting results, we know that

1. $V \subseteq B(V)$ (by the lemma).
2. If $W \subseteq B(W)$, then $B(W) \subseteq V$ (by theorem 1).
3. B is monotone and maps compact sets into compact sets.

Facts 1 and 2 imply that $V = B(V)$, so that the set of equilibrium values is a "fixed point" of B, in particular, the *largest* bounded fixed point. Monotonicity of B and the fact that it maps compact sets into compact sets provides an algorithm for computing the set V, namely, to start with a set W_0 for which $V \subseteq B(W_0) \subseteq W_0$, and to iterate on B. In more detail, we use the following steps:

1. Start with a set $W_0 = [\underline{w}_0, \overline{w}_0]$ that we know is bigger than V, and for which $B(W_0) \subseteq W_0$. It will always work to set $\overline{w}_0 = \max_{(x,y)\in C} r(x,y)$, $\underline{w}_0 = \min_{(x,y)\in C} r(x,y)$.
2. Compute the boundaries of the set $B(W_0) = [\underline{w}_1, \overline{w}_1]$. The value $\overline{w}_1$ solves the problem
$$\overline{w}_1 = \max_{(x,y)\in C} (1-\delta)\, r(x,y) + \delta \overline{w}_0$$
subject to
$$(1-\delta)\, r(x,y) + \delta \overline{w}_0 \geq (1-\delta)\, r(x,\eta) + \delta \underline{w}_0 \quad \text{for all} \quad \eta \in Y.$$
The value $\underline{w}_1$ solves the problem
$$\underline{w}_1 = \min_{(x,y)\in C;\, (w_1,w_2)\in[\underline{w}_0,\overline{w}_0]^2} (1-\delta)\, r(x,y) + \delta w_1$$
subject to
$$(1-\delta)\, r(x,y) + \delta w_1 \geq (1-\delta)\, r(x,\eta) + \delta w_2 \quad \forall\, \eta \in Y.$$
With $(\underline{w}_0, \overline{w}_0)$ chosen as before, it will be true that $B(W_0) \subseteq W_0$.
3. Having constructed $W_1 = B(W_0) \subseteq W_0$, continue to iterate, producing a decreasing sequence of compact sets $W_{j+1} = B(W_j) \subseteq W_j$. Iterate until the sets converge.

In section 24.13, we will present a direct way to compute the best and worst SPE values, one that evades having to iterate on the B operator.